Seokyung Park

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Research Interest | *Agricultural Robotics, Plant Phenotyping*

Education

Pusan National University, Busan, Republic of Korea Mar 2025 – Feb 2027 (expected)

- M.S. Bio-Industrial Machinery Engineering (GPA 4.36/4.50)
- Selected coursework: Biological Image Processing; Statistical Processing and Data Analysis; Advanced Bio-Industrial Machinery Design; Biosystems Mechanics

Pusan National University, Busan, Republic of Korea Mar 2017 – Feb 2025

- B.S. Plant Bioscience, Minor in Bio-Industrial Machinery Engineering (GPA 4.26/4.50)
- Summa Cum Laude; President's List
- Selected coursework: Biosensor and Bio-Instrumentation; Biosystems Control Engineering; Bio-Production Mechatronics Design; Biostatistics; Plant Physiology; Protected Horticulture

Research Experiences

Precision Agriculture and Smart Sensing System Lab., Pusan National University *Jun 2024 – Present*

Graduate Researcher; Undergraduate RA until Feb 2025. Advisor: Prof. Jaesung Park

- Tomato harvesting gripper: wrote Python code to run compression tests and process force signals on a myCobot 320 Pi with an adaptive gripper and force sensor, comparing four loading rates (6.9–15.3 N/s).
- Used hyperspectral images taken 1 h and 5 days after compression to separate immediate and delayed fruit damage; first-author manuscript under review at *Biosystems Engineering*.
- Greenhouse microclimate: measured temperature, relative humidity and CO_2 at two canopy heights in eight crop lanes with a Unitree Go2, and built 24 spatial maps to check how well the fixed sensors represent the house (co-first author).
- M.S. thesis: building a close-range 3D plant reconstruction workflow with a RealSense D456 on the Go2 (depth filtering, IMU/odometry pose correction, point-cloud registration, trajectory optimization), checked with overlap and nearest-neighbor distance metrics.
- Reservoir field survey: collected RGB images tagged with RTK-GNSS positions using the quadruped robot to build ground-surface datasets of upstream reservoir areas.
- Air-dome foundations: modeled expandable piles in Abaqus/Standard to see how wing angle and size change uplift load–displacement and soil plastic zones in sand; installed load cells on a dome in Damyang (Jun 2026) and logged cable tension through an indicator, RS-485 and a Raspberry Pi.

Functional Crop Laboratory, Pusan National University *Jan 2023 – Mar 2024*

Undergraduate Research Student

- Tracked soybean leaf traits under drought from time-series RGB images using YOLOv8 and image processing, with environmental data logged by Arduino-based sensors.

Selected Publications and Conferences

1. J. Park, **S. Park**, E. Jung, J. Song, G. Park, J. Lee, M. Kim and J. Park, "Inertial sensor applications in agriculture: Sensor types and data utilization techniques for mechanical and wearable applications," *Korean Journal of Agricultural Science*, 52(4), 447–469, 2025. doi:10.7744/kjoas.520405
2. **S. Park**, J. Song, H. Hwang, G. Park and J. Park, "Speed-dependent control of robotic gripper for tomato harvesting to minimize compression-induced damage," *Biosystems Engineering*, under review.
3. **S. Park**, J. Park, J. Jang, D. Park, S. Kwon, M. Kim and G. Choi, "Analysis of uplift resistance characteristics of expandable piles in sandy soil with varying wing inclination and dimensions," *Journal of the Korean Society*

of Agricultural Engineers, under review.

4. H. Hwang†, **S. Park**† and J. Park, "Greenhouse microclimate mapping using a quadruped robot to assess fixed sensor representativeness," in preparation. †Equal contribution.
5. **S. Park**, "Height-resolved 3D microclimate mapping and spatiotemporal analysis within greenhouse crop canopies using a quadruped robot," *ASABE Annual International Meeting*, Indianapolis, USA, Jul 2026 (AKABFE Student Research Spotlight).
6. **S. Park**, J. Park, H. Hwang and S. Kang, "Ground-reaction-force-based lateral slope estimation for furrow-following quadruped robots," *Korean Society for Agricultural Machinery Spring Conference*, 2026.
7. **S. Park**, J. Park, J. Song and M. Kim, "Analysis of pile pull-out behavior in Middle Eastern soil using digital image correlation," *Korean Society of Agricultural Engineers Autumn Conference*, Jeju, Oct 2025.

Selected Honors and Awards

Best Paper Presentation Award (team), *Korean Society for Precision Agriculture* *May 2025*

- Also received in Nov 2024

Excellent Paper Presentation Award, *Korean Society for Agricultural Machinery* *Apr 2025*

- Also received in Oct 2024

Excellence Award, *PNU–PKNU Field Problem Research Group Final Symposium* *Nov 2024*

Best Presentation Award (team), *Korean Society for Engineering Education* *Sep 2024*

Bronze Prize, *PNU College of Natural Resources and Life Science Capstone Design Competition* *Oct 2023*

Excellence Award, *Busan LINC 3.0 Entrepreneurship Shared University Competition* *Jan 2023*

Merit Scholarships, *Pusan National University* *2017 – 2024*

- Four full-tuition and two partial-tuition awards

Selected Projects

Quadruped Locomotion and Autonomous Control for Korean Farmland, *NRF / MOE* *Sep 2025 – Aug 2026*

- Principal investigator, NRF Master's Student Research Grant (RS-2025-25438074)

Quadruped-based Integrated Sensing for Agricultural Field Surveys, *NRF / MSIT* *Mar 2026 – Present*

- Participating researcher: sensor platform, greenhouse microclimate mapping, 3D phenotyping

Quadruped-based Upstream Soil Survey for Reservoir Embankments (Phase I), *NDMI* *Apr 2026 – Oct 2026*

- Participating researcher: robot field surveys with RGB imaging and RTK-GNSS

Low-energy Air-dome Vertical Farming for Hot Climates, *IPET / MAFRA* *Apr 2025 – Present*

- Participating researcher: foundation uplift analysis and structural load-cell sensing

Campus Mobility, Ride-sharing and Dining App Development, *Personal Project*

- Built a Dart/Flutter app for transport planning, real-time taxi ride-sharing, chat, fare settlement and dining; used Riverpod, GoRouter and Freezed with Supabase PostgreSQL, Auth, Realtime, Storage and Edge Functions.

Skills Summary

Programming: Python (NumPy, pandas, SciPy), C/C++

Robotics/Sensing: ROS 2, Unitree Go2, RTK-GNSS, LiDAR, IMU, RealSense D456

Instrumentation: Force sensors, load cells, Raspberry Pi, Arduino, RS-485

Imaging/Vision: RGB and hyperspectral imaging, YOLOv8, Segment Anything (SAM), point-cloud processing

App Development: Dart/Flutter, Riverpod, GoRouter, Freezed, Supabase/PostgreSQL

Simulation: Abaqus/CAE, Abaqus/Standard (VPK PLM Solution training courses, Mar and Jun 2026)